

From Working Code to Research Publication: Complete Workflow Guide

Overview: Your Research Pipeline

Your reorganized code creates a powerful research pipeline for analyzing ordinal categorical correlations. Here's how to use it for publication-quality research:

Real BES Data → Analysis → Insights → Publication

Step 1: Prepare Your Real BES Data

1.1 Load and Validate Your BES Dataset

```
r

# Load your actual BES data
bes_data <- read.csv("path/to/your/BES_2019_data.csv")

# Or if you have it in another format:
# bes_data <- readRDS("BES_2019_processed.rds")
# bes_data <- haven::read_dta("BES_2019.dta")

# Validate the data structure
cat("Dataset dimensions:", dim(bes_data), "\n")
cat("Required columns present:", all(c("var1", "var2", "corr", "nobs", "var1cats", "var2cats") %in
# Quick data overview
head(bes_data)
summary(bes_data[c("corr", "nobs", "var1cats", "var2cats")])
```

1.2 Data Quality Checks

```
r
```

```

# Create data quality report
source("utilities/data_loading.R") # If you created this file

# Comprehensive data validation
data_validation <- validate_bes_dataset(bes_data)
cat("Data validation passed:", data_validation$valid, "\n")
if (!data_validation$valid) {
  cat("Issues found:\n")
  for (error in data_validation$errors) {
    cat("  -", error, "\n")
  }
}

# Clean the data if needed
bes_data_clean <- clean_bes_data(bes_data)
cat("Cleaned dataset size:", nrow(bes_data_clean), "variable pairs\n")

```

Step 2: Run Your Research Analysis

2.1 Configure Analysis Parameters

```

r

# Set up research configuration
research_config <- list(
  nsim = 2000,      # High-quality simulations for publication
  matrix_trials = 200, # Comprehensive matrix testing
  min_vars = 5,     # Meaningful matrix sizes
  max_vars = 25,    # Computational feasibility
  progress = TRUE,  # Track long-running analysis
  seed = 42,        # Reproducibility
  significance_alpha = 0.05 # Standard significance level
)

cat("Research configuration:\n")
print(research_config)

```

2.2 Run Complete Analysis Pipeline

```
r

# Load the main analysis system
source("main_analysis.R")

# Run comprehensive analysis
cat("Starting research analysis on", nrow(bes_data_clean), "variable pairs...\n")
cat("This may take several minutes for large datasets.\n")

# Time the analysis for reporting
start_time <- Sys.time()
research_results <- run_complete_analysis(bes_data_clean, research_config)
end_time <- Sys.time()

analysis_time <- as.numeric(difftime(end_time, start_time, units = "mins"))
cat("Analysis completed in", round(analysis_time, 1), "minutes\n")
```

2.3 Generate Research Outputs

```
r

# Create output directories for publication
dir.create("research_output", showWarnings = FALSE)
dir.create("research_output/figures", showWarnings = FALSE)
dir.create("research_output/tables", showWarnings = FALSE)
dir.create("research_output/data", showWarnings = FALSE)

# Save complete results
saveRDS(research_results, "research_output/data/complete_analysis_results.rds")
cat("✓ Complete results saved\n")
```

Step 3: Generate Publication-Quality Figures

3.1 Main Research Figures


```
# Load visualization functions
```

```
source("1_bivariate_ordcats_correlation/3_visualization/bounds_visualization.R")
```

```
# Figure 1: Correlation Bounds Landscape
```

```
fig1 <- plot_bounds_scatter(research_results$bivariate$bounds_data, color_by = "bounds_rank")
```

```
fig1 <- fig1 +
```

```
  labs(title = "Theoretical Correlation Bounds for Ordinal Variables",
```

```
        subtitle = paste("British Election Study 2019 (n =", nrow(research_results$bivariate$bounds_data), ")")
```

```
  theme(text = element_text(size = 12), plot.title = element_text(size = 14, face = "bold"))
```

```
ggsave("research_output/figures/figure1_correlation_bounds_landscape.png",
```

```
       fig1, width = 10, height = 8, dpi = 300)
```

```
# Figure 2: Rescaling Effects
```

```
if (!is.null(research_results$rescaling$rescaled_data$r_rescaled)) {
```

```
  fig2 <- plot_rescaling_comparison(research_results$rescaling$rescaled_data)
```

```
  fig2 <- fig2 +
```

```
    labs(title = "Impact of Correlation Rescaling on Observed Relationships",
```

```
          subtitle = "Rescaling maps correlations to theoretical [-1,1] range") +
```

```
    theme(text = element_text(size = 12), plot.title = element_text(size = 14, face = "bold"))
```

```
ggsave("research_output/figures/figure2_rescaling_effects.png",
```

```
       fig2, width = 10, height = 8, dpi = 300)
```

```
}
```

```
# Figure 3: Asymmetry Analysis
```

```
if (!is.null(research_results$bivariate$asymmetry)) {
```

```
  asymmetry_data <- research_results$bivariate$asymmetry$detailed_results
```

```
  fig3 <- ggplot(asymmetry_data, aes(x = basic_asymmetry)) +
```

```
    geom_histogram(bins = 50, fill = "steelblue", alpha = 0.7, color = "white") +
```

```
    geom_vline(xintercept = 0, linetype = "dashed", color = "red", size = 1) +
```

```
    labs(title = "Distribution of Correlation Bounds Asymmetry",
```

```
          subtitle = "Asymmetry = r_max + r_min (perfect symmetry = 0)",
```

```
          x = "Bounds Asymmetry", y = "Number of Variable Pairs") +
```

```
    theme_minimal() +
```

```
    theme(text = element_text(size = 12), plot.title = element_text(size = 14, face = "bold"))
```

```
ggsave("research_output/figures/figure3_asymmetry_distribution.png",  
      fig3, width = 10, height = 6, dpi = 300)  
}  
  
cat("✓ Main figures saved to research_output/figures\n")
```

3.2 Supplementary Figures

r

Supplementary Figure A: Matrix Properties Comparison

```
if (!is.null(research_results$matrices)) {  
  matrix_comparison <- data.frame(  
    Property = c("Positive Semidefinite", "Invertible", "Valid Correlation Matrix"),  
    Original = c(research_results$matrices$summary_psd$orig_psd_percent,  
      research_results$matrices$summary_psd$orig_invertible_percent,  
      research_results$matrices$summary_psd$orig_valid_percent),  
    Rescaled = c(research_results$matrices$summary_psd$rescaled_psd_percent,  
      research_results$matrices$summary_psd$rescaled_invertible_percent,  
      research_results$matrices$summary_psd$rescaled_valid_percent)  
  )  
  
  matrix_comparison_long <- reshape2::melt(matrix_comparison, id.vars = "Property")  
  
  figS1 <- ggplot(matrix_comparison_long, aes(x = Property, y = value, fill = variable)) +  
    geom_bar(stat = "identity", position = "dodge") +  
    geom_text(aes(label = paste0(round(value, 1), "%")),  
      position = position_dodge(width = 0.9), vjust = -0.5) +  
    scale_fill_manual(values = c("Original" = "lightblue", "Rescaled" = "lightcoral"),  
      name = "Matrix Type") +  
    labs(title = "Matrix Properties: Original vs Rescaled Correlations",  
      subtitle = paste("Based on", research_results$matrices$summary_psd$n_trials, "random  
x = "", y = "Percentage of Matrices") +  
    theme_minimal() +  
    theme(text = element_text(size = 12), plot.title = element_text(size = 14, face = "bold"),  
      axis.text.x = element_text(angle = 45, hjust = 1))  
  
  ggsave("research_output/figures/figureS1_matrix_properties.png",  
    figS1, width = 12, height = 8, dpi = 300)  
}  
  
cat("✓ Supplementary figures saved\n")
```

Step 4: Generate Publication Tables

4.1 Summary Statistics Table

r

```
# Table 1: Descriptive Statistics
```

```
bounds_data <- research_results$bivariate$bounds_data
```

```
table1 <- data.frame(
```

```
  Statistic = c("Number of Variable Pairs", "Mean Correlation", "SD Correlation",  
    "Mean r_min", "SD r_min", "Mean r_max", "SD r_max",  
    "Mean Bounds Range", "SD Bounds Range",  
    "Mean Sample Size", "SD Sample Size"),
```

```
  Value = c(
```

```
    nrow(bounds_data),
```

```
    round(mean(bounds_data$observed_r, na.rm = TRUE), 3),
```

```
    round(sd(bounds_data$observed_r, na.rm = TRUE), 3),
```

```
    round(mean(bounds_data$r_min, na.rm = TRUE), 3),
```

```
    round(sd(bounds_data$r_min, na.rm = TRUE), 3),
```

```
    round(mean(bounds_data$r_max, na.rm = TRUE), 3),
```

```
    round(sd(bounds_data$r_max, na.rm = TRUE), 3),
```

```
    round(mean(bounds_data$r_max - bounds_data$r_min, na.rm = TRUE), 3),
```

```
    round(sd(bounds_data$r_max - bounds_data$r_min, na.rm = TRUE), 3),
```

```
    round(mean(bounds_data$n_obs, na.rm = TRUE), 0),
```

```
    round(sd(bounds_data$n_obs, na.rm = TRUE), 0)
```

```
)
```

```
)
```

```
write.csv(table1, "research_output/tables/table1_descriptive_statistics.csv", row.names = FALSE)
```

```
print(table1)
```

4.2 Key Findings Table

r

Table 2: Key Research Findings

```
if (!is.null(research_results$bivariate$asymmetry)) {  
  asymmetry_summary <- research_results$bivariate$asymmetry$summary  
  
  table2 <- data.frame(  
    Finding = c(  
      "Proportion with Symmetric Bounds",  
      "Proportion with Mild Asymmetry",  
      "Proportion with Moderate Asymmetry",  
      "Proportion with High Asymmetry",  
      "Mean Rescaling Effect",  
      "Proportion with Increased Magnitude",  
      "Matrix PSD Rate (Original)",  
      "Matrix PSD Rate (Rescaled)",  
      "Mean Condition Number (Original)",  
      "Mean Condition Number (Rescaled)"  
    ),  
    Value = c(  
      paste0(round(100 * asymmetry_summary$prop_symmetric, 1), "%"),  
      paste0(round(100 * asymmetry_summary$prop_mildly_asymmetric, 1), "%"),  
      paste0(round(100 * asymmetry_summary$prop_moderately_asymmetric, 1), "%"),  
      paste0(round(100 * asymmetry_summary$prop_highly_asymmetric, 1), "%"),  
      round(research_results$rescaling$effects$summary$mean_magnitude_change, 3),  
      paste0(round(100 * research_results$rescaling$effects$summary$prop_increased, 1), "%"),  
      paste0(round(research_results$matrices$summary_psd$orig_psd_percent, 1), "%"),  
      paste0(round(research_results$matrices$summary_psd$rescaled_psd_percent, 1), "%"),  
      round(research_results$matrices$summary_psd$orig_mean_condition, 2),  
      round(research_results$matrices$summary_psd$rescaled_mean_condition, 2)  
    )  
  )  
  
  write.csv(table2, "research_output/tables/table2_key_findings.csv", row.names = FALSE)  
  print(table2)  
}  
  
cat("✓ Tables saved to research_output/tables/\n")
```



Step 5: Generate Research Report

5.1 Automated Research Summary

r

Generate executive research summary

```
research_summary <- paste0(
  "CORRELATION BOUNDS ANALYSIS: BRITISH ELECTION STUDY 2019\n",
  "=" %&% strrep("=", 55), "\n\n",

  "DATASET OVERVIEW:\n",
  "- Variable pairs analyzed: ", nrow(bounds_data), "\n",
  "- Mean correlation: ", round(mean(bounds_data$observed_r, na.rm = TRUE), 3),
  " (SD = ", round(sd(bounds_data$observed_r, na.rm = TRUE), 3), ")\n",
  "- Mean sample size: ", round(mean(bounds_data$n_obs, na.rm = TRUE), 0), "\n\n",

  "KEY FINDINGS:\n",
  "1. Theoretical bounds range: [",
  round(mean(bounds_data$r_min, na.rm = TRUE), 3), ", ",
  round(mean(bounds_data$r_max, na.rm = TRUE), 3), "] on average\n",

  "2. Bounds asymmetry: ",
  if (!is.null(research_results$bivariate$asymmetry)) {
    paste0(round(100 * (1 - research_results$bivariate$asymmetry$summary$prop_symmetric),
      "% of pairs show asymmetric bounds")
  } else {"Analysis pending"}, "\n",

  "3. Rescaling effects: ",
  if (!is.null(research_results$rescaling)) {
    paste0(round(100 * research_results$rescaling$effects$summary$prop_increased, 1),
      "% of correlations increased in magnitude")
  } else {"Analysis pending"}, "\n",

  "4. Matrix properties: ",
  if (!is.null(research_results$matrices)) {
    paste0(round(research_results$matrices$summary_psd$orig_psd_percent, 1),
      "% of original matrices are positive semidefinite")
  } else {"Analysis pending"}, "\n\n",

  "IMPLICATIONS:\n",
  "- Correlation bounds vary substantially across variable pairs\n",
  "- Rescaling provides more interpretable correlation magnitudes\n",
```

```
"- Matrix properties are generally preserved under rescaling\n",  
"- Results support the use of bounds-aware correlation analysis\n\n",
```

```
"Generated: ", Sys.time(), "\n",
```

```
"Analysis time: ", round(analysis_time, 1), " minutes\n"
```

```
)
```

```
# Save research summary
```

```
writeLines(research_summary, "research_output/research_summary.txt")
```

```
cat("✓ Research summary saved\n")
```

```
cat("\nRESEARCH SUMMARY:\n")
```

```
cat(research_summary)
```

Step 6: Prepare for Publication

6.1 Methods Section Draft

r

```
# Generate methods section text
```

```
methods_text <- paste0(
```

```
"METHODS\n\n",
```

```
"Data Source:\n",
```

```
"We analyzed correlation data from the British Election Study 2019, examining ",  
nrow(bounds_data), " pairwise correlations between ordinal categorical variables.\n\n",
```

```
"Theoretical Framework:\n",
```

```
"For each pair of ordinal variables with fixed marginal distributions, we computed ",  
"theoretical correlation bounds using Fréchet-Hoeffding copula bounds. The minimum ",  
"possible correlation (r_min) was calculated using anti-comonotonic pairing, while ",  
"the maximum possible correlation (r_max) used comonotonic pairing.\n\n",
```

```
"Statistical Analysis:\n",
```

```
"1. Bounds Calculation: Theoretical bounds computed for all ", nrow(bounds_data), " variable p
```

```
"2. Permutation Testing: ", research_config$nsim, " permutation samples per pair for null distri
```

```
"3. Asymmetry Analysis: Bounds asymmetry measured as r_max + r_min\n",
```

```
"4. Rescaling Analysis: Linear rescaling mapping bounds to [-1,1] interval\n",
```

```
"5. Matrix Properties: ", research_config$matrix_trials, " random correlation matrix trials\n\n",
```

```
"All analyses conducted in R version ", R.version.string, ".\n"
```

```
)
```

```
writeLines(methods_text, "research_output/methods_section_draft.txt")
```

```
cat("✓ Methods section draft saved\n")
```

6.2 Results Section Key Points

r

```
# Generate results bullet points
```

```
results_bullets <- c(  
  paste0("• Analyzed ", nrow(bounds_data), " ordinal variable correlations from BES 2019"),  
  paste0("• Mean observed correlation: ", round(mean(bounds_data$observed_r, na.rm = TRUE),  
    " (range: ", round(min(bounds_data$observed_r, na.rm = TRUE), 3), " to ",  
    round(max(bounds_data$observed_r, na.rm = TRUE), 3), ")"),  
  paste0("• Mean theoretical bounds: [", round(mean(bounds_data$r_min, na.rm = TRUE), 3),  
    ", ", round(mean(bounds_data$r_max, na.rm = TRUE), 3), "]",  
  
  if (!is.null(research_results$bivariate$asymmetry)) {  
    paste0("• Bounds asymmetry: ",  
      round(100 * (1 - research_results$bivariate$asymmetry$summary$prop_symmetric), 1),  
      "% of pairs showed non-symmetric bounds")  
  },  
  
  if (!is.null(research_results$rescaling)) {  
    paste0("• Rescaling effects: ",  
      round(100 * research_results$rescaling$effects$summary$prop_increased, 1),  
      "% of correlations increased in magnitude after rescaling")  
  },  
  
  if (!is.null(research_results$matrices)) {  
    paste0("• Matrix validity: ",  
      round(research_results$matrices$summary_psd$orig_psd_percent, 1),  
      "% of correlation matrices were positive semidefinite")  
  }  
)
```

```
# Remove NULL entries
```

```
results_bullets <- results_bullets[!sapply(results_bullets, is.null)]
```

```
writeLines(results_bullets, "research_output/results_key_points.txt")
```

```
cat("✓ Results key points saved\n")
```

```
for (bullet in results_bullets) {
```

```
cat(bullet, "\n")
}
```

Step 7: Next Research Steps

7.1 Reproducibility Package

```
r

# Create reproducibility information
repro_info <- list(
  analysis_date = Sys.time(),
  r_version = R.version.string,
  platform = R.version$platform,
  config = research_config,
  data_summary = list(
    n_pairs = nrow(bounds_data),
    date_analyzed = Sys.Date()
  ),
  package_versions = sessionInfo()
)

saveRDS(repro_info, "research_output/data/reproducibility_info.rds")
cat("✓ Reproducibility information saved\n")
```

7.2 Future Analysis Directions

```
r
```

Document potential extensions

```
future_directions <- c(
  "1. Cross-validation with other survey datasets (e.g., ANES, ESS)",
  "2. Comparison with polychoric correlation estimates",
  "3. Impact of rescaling on multivariate statistical models",
  "4. Bounds-aware correlation network analysis",
  "5. Extension to mixed-type variable correlations",
  "6. Temporal stability of bounds across BES waves",
  "7. Machine learning applications with rescaled correlations"
)

writeLines(future_directions, "research_output/future_research_directions.txt")
cat("✓ Future directions documented\n")
```

Final Research Output Summary

You now have a complete research analysis package:

research_output/

```
├── figures/
│   ├── figure1_correlation_bounds_landscape.png
│   ├── figure2_rescaling_effects.png
│   ├── figure3_asymmetry_distribution.png
│   └── figureS1_matrix_properties.png
├── tables/
│   ├── table1_descriptive_statistics.csv
│   └── table2_key_findings.csv
├── data/
│   ├── complete_analysis_results.rds
│   └── reproducibility_info.rds
├── research_summary.txt
├── methods_section_draft.txt
├── results_key_points.txt
└── future_research_directions.txt
```


Congratulations!

You've successfully transformed your R code into a complete research pipeline that produces publication-ready outputs. Your analysis is:

-  **Methodologically sound** - Based on rigorous correlation bounds theory
-  **Computationally robust** - Handles large datasets efficiently
-  **Publication ready** - Generates high-quality figures and tables
-  **Reproducible** - Complete documentation and configuration tracking
-  **Extensible** - Modular structure supports future research

Next immediate steps for publication:

1. Run this workflow on your complete BES dataset
2. Review and refine the generated figures and tables
3. Integrate the methods and results text into your manuscript
4. Consider the future research directions for follow-up studies

Your reorganized code has successfully evolved from a collection of scripts into a powerful research instrument! 